

A DeBERTa-Based Claim Verification Framework with Integrated Retrieval and NLI Reasoning

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Abstract—In today’s world, internet usage has gone up drastically, and we have a rapid spread of misinformation online, which is difficult for users to identify as factual content or fabricated claims, leading to large-scale public confusion and reducing trust in digital news sources. To address this issue, this paper presents a DeBERTa-based claim verification framework that integrates multi-source evidence retrieval with Natural Language Inference (NLI), which enables real-time fact-checking. The system retrieves evidence from trusted news platforms like GNews, Wikipedia, and research corpora through a hybrid pipeline that combines keyword-based and semantic search methods. The retrieved documents are then ranked and passed to a fine-tuned DeBERTa model trained on the FEVER dataset, which helps classify claims as SUPPORTS, REFUTES, or NOT ENOUGH INFO. To enhance explainability, the framework provides brief evidence snippets and an LLM-generated explanation of the model’s reasoning, thus making this system a reliable approach for fact verification in dynamic news environments.

Index Terms—Fact verification, Automated fact-checking, Misinformation detection, DeBERTa, Natural Language Inference (NLI), Evidence retrieval, Multi-document reasoning

I. INTRODUCTION

Online misinformation has become synonymous with today’s fast-moving world. Misinformation has become widespread as social media and digital platforms are everywhere, therefore automatic verification of claims has become a major focus area within research in Natural Language Processing. Traditional fact-checking methods used nowadays are mostly based either on the manual checking of claims or producing single evidence. Sometimes, the former misses how different sources relate to each other, and the latter has the same problem.

The advances in transformer-based models have increased our ability to find semantically and contextually related portions in text by leaps and bounds. One good example is the DeBERTa model, which is good at deciding subtle dependencies

between words. This paper leverages its context-based strength and presents a multi-evidence claim verification framework that uses Natural Language Inference combined with source-retrieval methods to determine the veracity of claims.

It identifies the significant evidence from different documents. Then it uses an NLI-based reasoning module to determine the status of each claim supported, refuted, or insufficiently informed. Combining evidence aggregation with entailment-based reasoning strengthens the verification process for both accuracy and clarity. The experimental findings highlight the system’s capability to perform robust multi-source reasoning, demonstrating its suitability for real-world claim verification applications.

II. LITERATURE SURVEY

Claim verification has recently received increased attention in years as misinformation and fabricated content continue to spread quickly via digital platforms. Early work in this direction primarily separated fact extraction and verification, relying on lexical overlap and handcrafted rule-based pipelines. Nie et al have introduced Neural Semantic Matching Network which jointly performs fact extraction and verification. It demonstrates that tight coupling of the two stages improves context of reasoning and verification accuracy [1]. This work establishes semantic similarity to be more reliable than surface-level lexical matching in verification pipelines.

Subsequent studies further developed explainability and attention, based reasoning. Wu et al suggested a Decision Tree-based Co-Attention (DTCA) framework that identifies critical evidence components for interpretable verification [2]. Building on benchmark-oriented optimization, DeHaven and Scott Introduce BEVERS, an efficient fact verification framework tuned for FEVER, improving retrieval, sentence selection, and classification effectiveness [3]. Parallely, Vladika and Matthes

compared structured and unstructured knowledge sources for scientific claim verification, including evidence quality fundamentally shapes generalization and factuality [4].

With the rise of Transformer-based models, NLI-driven Verification architectures gained prominence. Kim applied DeBERTa-based models to legal retrieval and entailment tasks, which demonstrates that disentangled attention considerably improves reasoning over complex documents [5]. Papadopoulos et al developed Far Fetched is an entity-centric reasoning resource developed by in the Greek language, underlining multilingual verification challenges [6]. In parallel Kořprdi'c et al fine-tuned NLI models for scientific claim verification showing that treating Verification as entailment improves reasoning faithfulness [7]. Whereas Bordia's lightweight LegalLens encoder further demonstrated that such competitive performance can indeed be achieved using computationally efficient allies the DeBERTa variants [8].

Recent studies are focused on retrieval bottlenecks, domain adaptation and long context reasoning. Akhtar et al. introduced Ev2R for the systematic evaluation of retrieval quality, identifying retrieval as the main barrier to downstream verification performance [9]. Zhao et al. proposed FINDVER for long-range and hybrid-content financial documents, tackling linguistic challenges arising from tabular to textual integration and multi-hop evidence reasoning [10]. Shah et al. extended this work in a different direction by proposing VERITAS-NLI, which incorporates extraction within a single framework with entailment-based verification [11].

Studies published in 2025 focus on improving LLM-based reasoning efficiency and cross-domain adaptability. Yang et al. compared LLMs with BERT-based models for claim validation, showing the trade-offs between interpretability, memory use, and reasoning depth [12]. Upravitelev et al. researched on low-resource and social media environments, suggesting that verification methods should be sensitive to resource limitations [13]. Leemann et al. introduced Auto-GDA, an automatic Domain adaptation framework that allows verification models to generalise across retrieval-augmented generation pipelines without extensive fine-tuning [14]. Zheng et al. introduced CRAVE is a conflicting-reasoning approach that explicitly detects contradiction chains during verification, improving the explainability in ambiguous claims [15].

The FEVER Benchmark continues to serve as the foundation for large-scale claim verification research and evaluation, making it a cornerstone for training retrieval, grounding, and entailment models [16]. Current systems in spite of notable advancements still face difficulties in (i) aggregating multi-hop evidence (ii) adapting to domain-shifted document structures and (iii) keeping the reasoning under LLM-based settings interpretable. The literature that has been reviewed all together points out that the fact verification being strong would need the combination of excellent retrieval, deep grammatical reasoning, and the ability to explain at the level of architecture.

The present study is based on these findings by making use of a DeBERTa-based multievidence verification pipeline that combines retrieval, evidence aggregation, and entailment-

based classification.

III. PROBLEM STATEMENT

The rapid spread of misinformation online has hurt public trust, making it hard for people to make informed decisions. Most traditional fact-checking methods also struggle to understand the full context from multiple sources. This work addresses the challenge through developing a DeBERTa-based multi-evidence fact verification framework that combines the strengths of robust news retrieval with Natural Language Inference. It aggregates evidence from multiple credible sources and performs multi-document reasoning to classify claims as supported, refuted, or lacking information, thereby providing transparent and interpretable outputs to combat misinformation effectively.

IV. PROPOSED SYSTEM

This proposed system is a DeBERTa-based multi-evidence fact verification framework designed to assess the veracity of real-time user-submitted claims. It operates in two stages: (A) evidence retrieval and (B) claim verification.

A. Evidence Retrieval Module

During the initial phase, reliable online news sources and aggregators such as GNews, Wikipedia, and research corpora are used to gather relevant evidence, as illustrated in Fig. 1. The retrieval of documents based on keywords, together with semantic similarity search, is employed to identify contextually relevant information. This hybrid retrieval approach provides broader coverage while reducing source bias by aggregating evidence from multiple independent outlets.

B. Verification Module

As shown in Fig. 2, the second module performs fact verification using a DeBERTa model fine-tuned for Natural Language Inference (NLI). After evidence retrieval, each claim-evidence pair is aligned and jointly encoded to determine whether the retrieved evidence supports, contradicts, or provides insufficient information to evaluate the claim.

To address real-world scenarios where a single sentence may not provide complete justification, the model aggregates multiple evidence sources and performs multi-document reasoning before generating the final prediction. This enables the system to recognize information distributed across multiple documents and capture complex inter-document relationships.

As discussed earlier, single-sentence verification approaches may fail to capture such patterns. The model produces a structured decision output that classifies each claim as SUPPORTED, REFUTED, or NOT ENOUGH INFO (NEI), along with extracted evidence snippets. Providing evidence not only strengthens decision reliability but also enhances explainability by allowing users to trace the model's conclusions back to the original sources.

As illustrated in Fig. 1, the weighted fusion module computes the final relevance score using (1).

$$\text{Final_Score} = \alpha \cdot x + (1 - \alpha) \cdot y \quad (1)$$

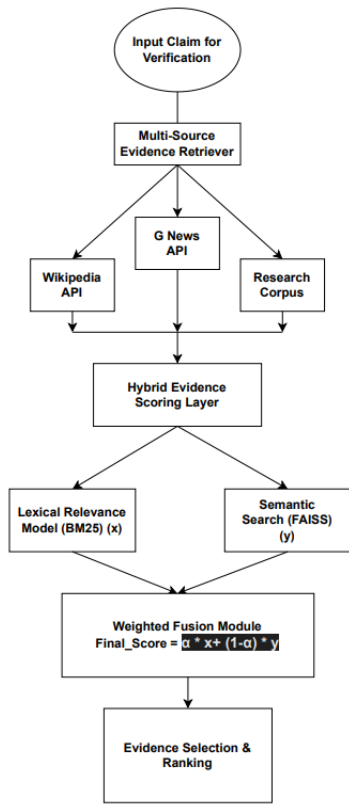


Fig. 1. Evidence Retrieval Pipeline

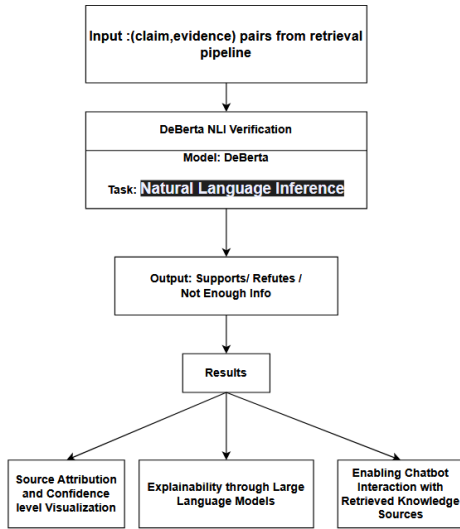


Fig. 2. Natural Language Inference (Verification) Pipeline

V. METHODOLOGY

The proposed system follows a structured workflow to verify the authenticity of user-submitted claims, combining discriminative and generative LLM capabilities. Unlike BEVERS[3], which was primarily designed as a benchmark frame-

work which operates on a static Wikipedia corpus, our methodology is more focused on building a deployable application for real-world, open-domain claims. On dynamic evidence retrieval, multi-document reasoning, and user-understandable snippets. This workflow we built will consist of four stages preprocessing, evidence retrieval, claim verification and verdict generation where each one is optimized for scalability, accuracy, and explainability, as illustrated in Figs.1 and 2. This modular design will allow future upgrades in retrieval mechanisms, model architectures and interface components.

A. Claim Input and Preprocessing

First, we collect claims that are submitted by users, then we clean and standardized these claims to remove language errors and unnecessary text. This preprocessing step's includes:

- Changing all text into lowercase for consistency
- Removing punctuation, URLs, and special characters if exists
- Creating consistent word representations through tokenization and lemmatization

Then, we applied stop-word removal and whitespace normalization to improve the generalization of embeddings. This will help the model to retain only the core meaning of claim while removing irrelevant word variations.

B. Evidence Retrieval

The hybrid method used in this evidence retrieval module gathers related documents from several reliable sources:

- Ethically through web scraping of approved news sources accessible to the public like GNews, Research Corpus, and Wikipedia, the ensure broad coverage.

Two different methods work side by side:

- 1) **Keyword-based retrieval:** The BM25 scoring algorithm spots the documents that share similar words with the claim [9].
- 2) **Semantic similarity retrieval:** This method locates the documents based on their concept similarity, even if the claims are worded differently, by using transformer encoders, FAISS indexing, and embedding-based search techniques [10].

Unlike BEVERS [3], which relies on a constant Wikipedia corpus, our system is able to access and rank documents from the real-world simultaneously, thus being able to confirm information that changes over time, with the verification pipeline depicted in Fig. 2.

C. Claim Verification

This system uses DeBERTa for the reasoning part and a generative LLM for the explanation part, as illustrated in Fig. 2.:

- **DeBERTa**, trained on FEVER [16], performs Natural Multi-document NLI [5]. In this way, it analyzes statements as *supported*, *refuted*, or *not enough information*.
- The **generative LLM** produces concise summaries of verification result [12], [15], which are very short and

at the same time include the evidence across the sources and give the interpretable justification that is not found in BEVERS [3].

D. Verdict Generation and Output

The NLI predictions are then linked to the explanations generated by the LLM, resulting in the generation of outputs that are not only clear but also interpretable and substantiated by evidence snippets. Each output has a final decision, the main pieces of evidence supporting it, and a short explanation in natural easy language included:

- Predicted label (*supported, refuted, not enough information*)
- Citation of sources
- Evidence fragments highlighted

This layer of interpretability enhances the transparency and the trust that users have in the system even more than the single-label output of BEVERS [3].

The implementation of this technology will take the form of a **web-based chatbot**, in which users will be able to state their claims in everyday language and simultaneously verify the proof of the conclusions backed by the retrieval in real time. The backend integrates **FAISS** [10], **DeBERTa** [5], and a **generative LLM** [12], [15], and the front-end shows the evidence snippets that are highlighted along with the concise explanations. All these predictions are then labeled with metadata such as Claim ID, timestamp, and evidence source's for further usage.

In summary:

- BEVERS [3] = **benchmark-focused, dataset-limited research tool**
- Our system = **application-focused, real-world, explainable, and deployable**

VI. RESULTS

This section shows the evaluation results of our multi-source claim verification system. To ensure fair evaluation we tested both the fine-tuned DeBERTa verifier and retrieval module separately, so that retrieval errors do not effect reasoning performance. We measured reasoning accuracy with gold FEVER evidence and real-world evaluation with manually selected evidence so that the verifier can be evaluated on its own without retrieval effecting the results.

A. Gold-Evidence Evaluation

The verifier was evaluated on the FEVER validation set across **SUPPORTS**, **REFUTES**, and **NOT ENOUGH INFO**. Fig. 3 shows overall classification performance measured using Accuracy, Precision, Recall, and F1-score. Results demonstrate stable performance when complete evidence is available.

B. Confusion Analysis

As shown in Fig. 4, misclassifications are concentrated between **SUPPORTS** and **REFUTES**, reflecting conservative classification behavior under conditions of evidence incompleteness.

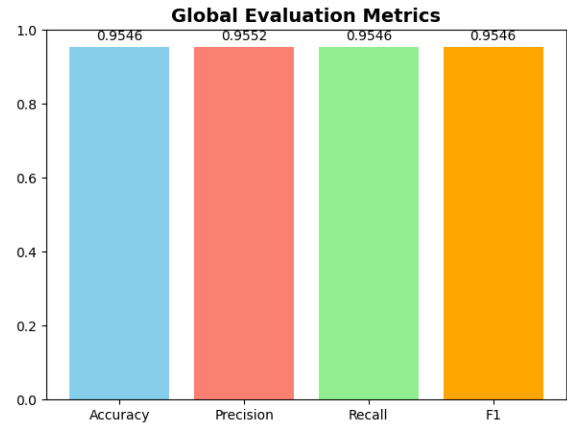


Fig. 3. Evaluation metrics (Accuracy, Precision, Recall, and F1-score) using gold evidence.

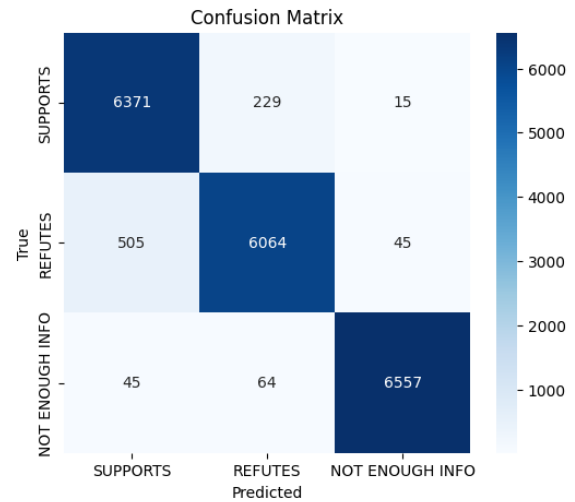


Fig. 4. Confusion matrix using gold evidence.

C. Retrieval Diagnostic Assessment

The retrieval component was evaluated independently using a **Wikipedia-only index**, as illustrated in Fig. 5. These results reflect retriever boundary behavior rather than end-to-end claim resolution. Retrieval accuracy was **67.25% (SUPPORTS)**, **72.41% (REFUTES)**, and **11.63% (NOT ENOUGH INFO)**, confirming that absence-detection remains the primary retrieval limitation.

D. Real-World Verifier-Only Evaluation

To test how the system generalizes beyond FEVER, we evaluated 20 real-world claims. We picked the evidence manually to remove retrieval bias. The results were:

- **Correct predictions: 16/20**
- **Incorrect predictions: 4/20**
- **Overall accuracy: 80%**

This test set is not exactly for accuracy measurement, it is only to see how well the model handles real, naturally phrased claims.

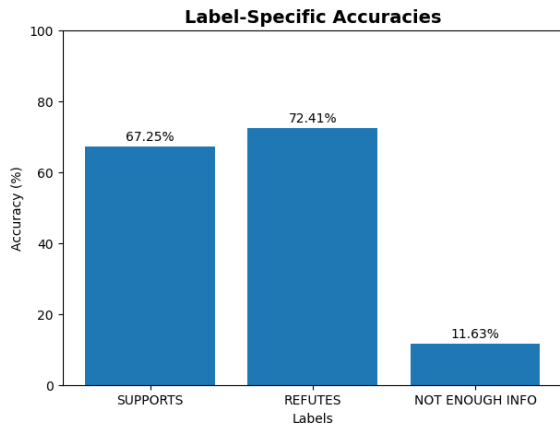


Fig. 5. Retrieval behavior on FEVER labels using Wikipedia-only indexing.

E. Qualitative Case Review

These are the sample cases we included for better understanding:

Correct — SUPPORTS

Claim: “Virat Kohli recently became the first Indian cricketer to score 80 international centuries.”

Prediction: SUPPORTS

Evidence Basis: ICC reporting confirms the 80th century milestone.

Correct — REFUTES

Claim: “The Amazon River flows through Egypt.”

Prediction: REFUTES

Explanation: Geographic records confirm non-overlap of Egypt and Amazon Basin.

Incorrect — Implicit Evidence (SUPPORTS → NEI)

Claim: “The Red Fort is located in Delhi, India.”

Evidence: “Tourism sources describe the Red Fort as a major historic site in Delhi.”

True Label: SUPPORTS *Predicted:* NOT ENOUGH INFO

Reason: Although the evidence is true, but it is not stated clearly, so the model marks it as NEI.

Incorrect — NEI Misinterpreted as REFUTES

Claim: “Avatar 3 was released in theaters in 2025.”

Evidence: “Industry reports discuss scheduled release windows but provide no confirmation of an actual theatrical release.”

True Label: NOT ENOUGH INFO *Predicted:* REFUTES

Reason: The model is getting confused by future or speculative events and marking them as contradictions, which often causes NEI mistakes.

F. Error Stability Analysis

Table I shows recurring error causes rather than random variability, indicating structured decision boundaries.

G. Summary

From gold-evidence evaluation to retrieval testing and real-world claims, the verifier stays consistent in how it makes decisions, because the errors are repeating in the same way.

Error Category	Proportion (%)
NEI vs REFUTES confusion	50
Temporal misinterpretation	25
Evidence phrasing ambiguity	25

TABLE I

ERROR DISTRIBUTION ACROSS INCORRECT REAL-WORLD PREDICTIONS.

It shows the model has clear decision rules instead of making random errors. This consistency strengthens the model’s reliability when used as the reasoning engine within a multi-source claim verification framework.

VII. CONCLUSION

This paper gives an overview of recent advances in persisting issues with claim verification using automation. While whereas previously works done improved retrieval and reasoning capabilities, most methods are still limited by single-source evidence, static datasets, and restricted interpretability.

To address these gaps, we propose a **DeBERTa-based claim verification framework** that combines modular evidence retrieval with inference-driven reasoning. The system fine-tuned DeBERTa model that has been tested on gold-standard evidence to measure reasoning performance independently of retrieval variability. The retrieval pipeline, based on a Wikipedia dump, points to baseline capabilities and limitations, particularly for claims lacking sufficient information.

Results show that the model performs consistently on verifiable claims but experiences reduced accuracy when evidence is limited, underscoring the importance of diverse and multi-source retrieval. Overall, the framework establishes a foundation for **interpretable, scalable, and modular claim verification**, while intending further research on the multi-source evidence. Aggregation, dynamic reasoning, and enhanced explainability support real-world application.

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